



WANG QUAN

Junior AI Application / Full-Stack Developer

Computer Science graduate focused on AI application and full-stack development. Built a production WeChat mini program for course booking and verification, and developed an AI digital human creation platform for voice cloning, media management, and lip-sync video workflows.

Portfolio

Detailed project write-ups, code links, and demo videos are on my portfolio site.

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Singapore | LTVP/PLOC work eligible

GitHub: github.com/wqmumu
Portfolio: wang-quan-portfolio-site.vercel.app
YouTube: [YouTube project demos](#)

CORE STACK

Frontend	React, TypeScript, Next.js, Vite, WeChat Mini Program, Tailwind CSS, shadcn/ui
Backend	Node.js, Express, REST APIs, validation, task polling, transaction-aware business logic
Database	MySQL, SQL migrations, Excel import workflows, eligibility data modeling
AI / Media	Voice cloning workflows, lip-sync video generation, RunningHub OpenAPI, media upload flows
Deployment	Git, GitHub, Vercel, npm/pnpm, PM2, Tencent Cloud deployment notes

SELECTED PROJECTS

TCM Course Booking & Verification Mini Program

Production v1.0 released | WeChat Mini Program, React, Node.js, MySQL

- Built a full booking ecosystem with a WeChat mini program, React/TypeScript admin dashboard, Node.js API, and MySQL database for course reservation and staff verification workflows.
- Implemented login, phone binding, purchase eligibility checks, purchased-course-only display, booking, capacity control, waitlist handling, and QR-code based verification.
- Developed admin workflows for course/session management, appointment and waitlist records, verifier accounts, Excel purchase-list import, import exceptions, student eligibility, and system settings.
- Supported production operation with deployed API/admin domains, Tencent Cloud server deployment, MySQL migrations, PM2 restart flow, and production data safeguards.

AI Digital Human Creation Platform

Portfolio project | Next.js, TypeScript, RunningHub OpenAPI, Tailwind CSS

- Developed a full-stack AI media workspace for creating digital human content from uploaded voice and portrait materials.
- Built voice synthesis features that accept script text and reference audio, call a voice-cloning workflow, and return generated audio for preview and download.
- Built video synthesis features that combine a selected portrait image with generated or uploaded audio to create lip-sync digital human video output.
- Implemented material library upload/list/delete flows, audio playback, image preview, task timers, cancellation, polling for RunningHub outputs, and file validation.

ADDITIONAL PROJECTS

Selected Coursework Project Gallery

Public GitHub repositories and YouTube demos | [React Native](#), [Unity](#), [Python](#), [C++](#), [p5.js](#)

- Organized 20 cleaned coursework repositories, separating formal projects from early exercises and excluding reports, large files, generated caches, assignment documents, and private data.
- Prepared selected public GitHub repositories and YouTube demo videos covering mobile, Unity game, VR, database/web, NLP, C++/JUICE audio, and p5.js creative coding work.

ThriftMate Mobile App

[Expo](#), [React Native](#), [React Navigation](#)

- Built a mobile coursework app with browse, detail, and favorites flows plus reusable UI structure.

Unity Gate Multiplier Game

[Unity 6](#), [C#](#), [URP](#), [Input System](#)

- Built a hyper-casual game prototype with procedural gates, troop pooling, formation logic, enemy combat, UI, and audio systems.

IMDB Sentiment Analysis

[Python](#), [Jupyter Notebook](#), [NLP](#)

- Completed an NLP coursework notebook for movie-review sentiment analysis and model-result explanation.

C++ / Creative Coding Coursework

[C++](#), [JUICE](#), [JavaScript](#), [p5.js](#), [matter.js](#)

- Completed supporting projects including a JUICE DJ/audio app, OOP data-processing projects, interactive graphics demos, and recorded walkthrough videos.

EDUCATION

Bachelor Degree in Computer Science

University of London | Jan 2023 - Oct 2025

Diploma in Information Technology

Singapore Institute of Management | Jun 2021 - Aug 2022

Links

github.com/wqmumu | wang-quan-portfolio-site.vercel.app | [YouTube demos](#)